

Questions			Answers	Marks
5	(a)		Angular velocity is the <u>rate of change of angular displacement</u> . Unit is rad s^{-1}	[B1] [B1]
	(b)		From forces p.o.v., $mg + T = \frac{mv^2}{L}$ Analyzing FBD of string just taut, when the particle reaches C Hence $T = 0 \text{ N}$ $mg = \frac{mv^2}{L}$ $v = \sqrt{gL}$	[M1] [B1] [A0]
	(c)		From energy p.o.v., initial KE = Gain in GPE + final KE $\frac{1}{2}mV^2 = mg2L + \frac{1}{2}mv^2$ $\frac{1}{2}mV^2 = mg2L + \frac{1}{2}mgL$ $V^2 = g4L + gL$ $V^2 = g4L + gL$ $V = \sqrt{5gL} = 7.0$	[M1] [A1]